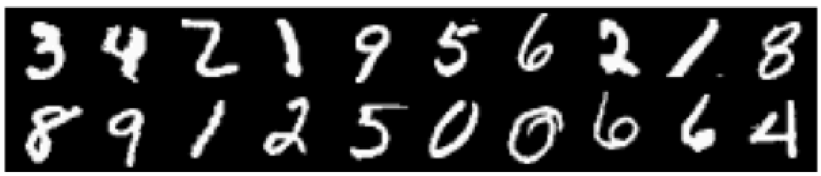
CS148a Assignment 1a: Adversarial MNIST Data Collection

In this assignment, you will create a small, human-interpretable dataset of ***in-the-wild numerical digits*** designed to be challenging for machine learning models. The goal is to experience data collection firsthand. You and your classmates will later build models to classify your own datasets.

Since this is a super easy assignment, we recommend that you do not use your late days for this assignment!

Background: What is MNIST?



MNIST (Modified National Institute of Standards and Technology) is a classic benchmark dataset consisting of 70,000 grayscale images of handwritten digits (0–9), each cropped to 28×28 pixels. It played a foundational role in the rise of modern machine learning by providing a standardized, accessible task for evaluating classification algorithms. Read more about its history on [Wikipedia](https://en.wikipedia.org/wiki/MNIST_database).

Task Overview

Each student will collect a personalized set of "adversarial MNIST" images: digits that are clearly recognizable to humans but intentionally difficult for standard models trained on MNIST-like data. They are adversarial because they come from ***out-of-distribution settings***.

Data Collection Requirements

- Create images of all ten digits: 0 through 9.
- Choose ***ONE distinct and "unlikely" writing or construction setting*** (e.g., branch-in-sand, crayon-on-wood, chalk-on-brick, letters-made-of-sphagetti, rainbow-origami-letters, etc.).
- For each digit, you should create 10 versions of images.
- This results in 10 digits × 1 setting × 10 versions = 100 images total.

For each of the five settings, give the setting a short descriptive name using hyphens (e.g., candy-arrangement, leaf-sticks, shadow-cast). Settings do not need to be wildly different, but should introduce meaningful variation in texture, geometry, lighting, or material.

Image Formatting and Naming

Each image must be tightly cropped so that the digit fits clearly within a frame. They should be *at least* 128x128px in dimension, capped at 1024x1024px. Note that they should be distinguishable by a human at your submitted resolution. Images should be saved using the following naming convention:

{firstname_lastname}/{setting}/{digit}_v{version}.jpg

Example:
I zip and submit *raphi_kang.zip*, inside of which is *spaghetti-letters*, which contains *3_v1.jpg*, *3_v2.jpg*, etc.

…

Design Philosophy

Your data will ultimately be evaluated against models trained by other students. You are therefore ***incentivized to create data that is maximally challenging for machine learning models***.

On the other hand, in the next assignment, three other students in the class will be assigned to label your dataset. You are thus ***incentivized to create data that is distinctly classifiable for humans***. The labelers will also rate how interesting/creative your settings are, and ***flag any AI-generated images***.

There will be a small award for the *coolest* datasets.

Use of AI Tools

While we encourage AI usage for the later parts of this project, YOU MAY NOT SUBMIT AI GENERATED IMAGES.

If you otherwise use AI in this project (to label your images for example), you must clearly document how they were used and include relevant conversation logs in your writeup, under an "AI usage" heading.

Student Deliverables

- A single ZIP folder containing all 100 images.
- A short PDF writeup describing:
  - Each of the five settings with two examples per setting
  - How digits were constructed and cropped
  - What differs between the two versions of each digit
  - Why you think your data will be harder than MNIST for an AI model
  - Any use of AI tools, with attached conversation logs
  - 1 selfie/video of yourself collecting the images in that setting:)

Release Date: 1/20/2026
Due Date: 1/27/2026